SEVEN TECHNIQUES

Measured against one baseline / SAM 2 and SAM 3

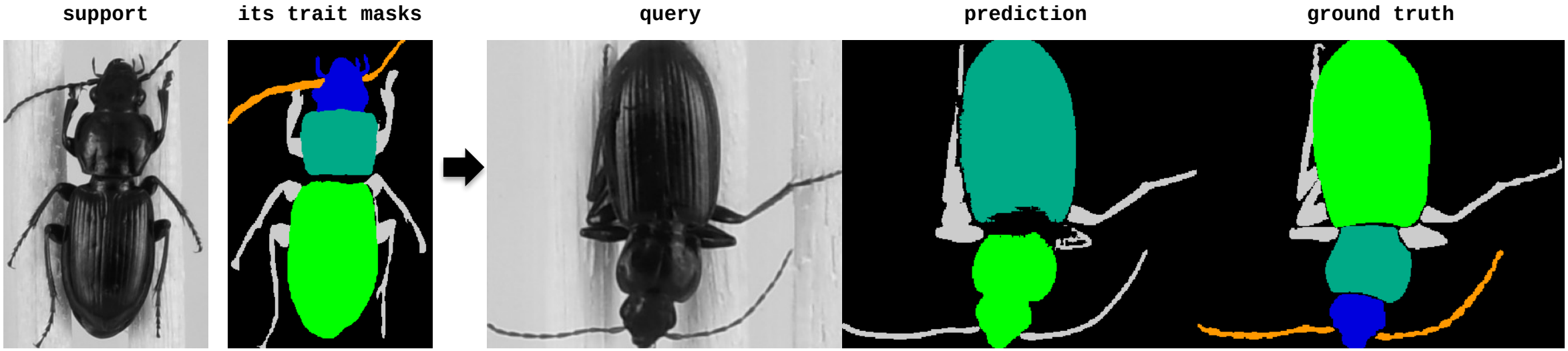
Support-side curation

Orientation TTA

Loss and model swaps

A findings talk / about 20 minutes / Hunter Bridges

The task



one 2-frame video · masks propagate from frame 0 to frame 1

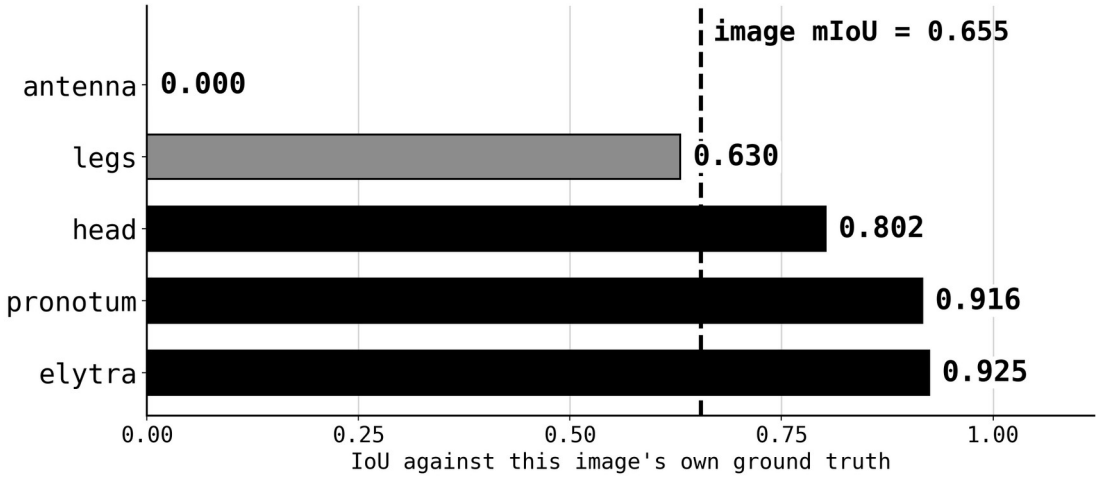
the query is pinned 180°, and the prediction follows the support's layout

THE UNIT EVERY NUMBER IS IN

How a score is computed

1. resize logits to GT resolution
2. each pixel takes its best-scoring trait
3. best logit <= 0 means background
4. IoU per trait present in this GT
5. image score = mean over those traits

one real query: beetle_1166_231, run 0



what a number means

1.00 is every pixel of every trait correct.

Traits are averaged, not pixels, so a hair-thin antenna counts as much as the elytra.

A ground-truth trait the model never claims scores 0.

then the mean over the images in a run, and the mean over 20 runs

THE PREMISE

The baseline

55.96 ± 7.90

Beetle, 1-shot, uniform supports

target: 61.9 ± 3.7

68.54 ± 3.15

Butterfly Major, 1-shot, uniform supports

target: 81.0 ± 1.0

how each technique is judged

20 draws / same draws both arms / paired per seed / CI / band frozen first

run-to-run spread is reported as a result, not an error bar

TECHNIQUE 1 · SUPPORT-SIDE

Curating the reference

01	**Measure the body axis**	directed vector between two trait centroids
02	**Take the threshold from the data**	the train pool's own circular std, 55.6°, not tuned
03	**Estimate on train only**	no test orientation may touch the pool
04	**Require every trait present**	the second half of the criterion

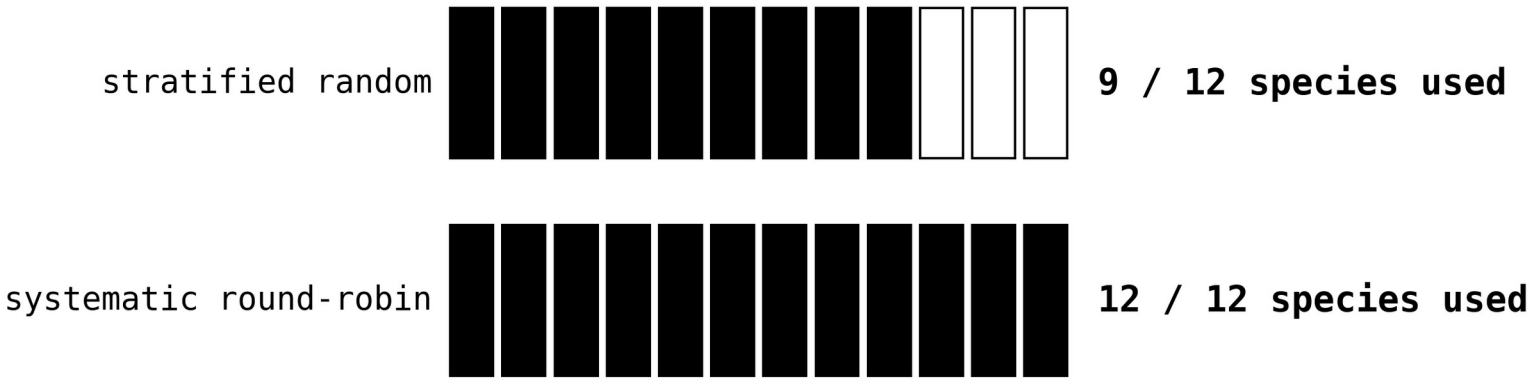
120 candidate references → a pool of 79

TECHNIQUE 2 · SUPPORT-SIDE

Fixing the draw

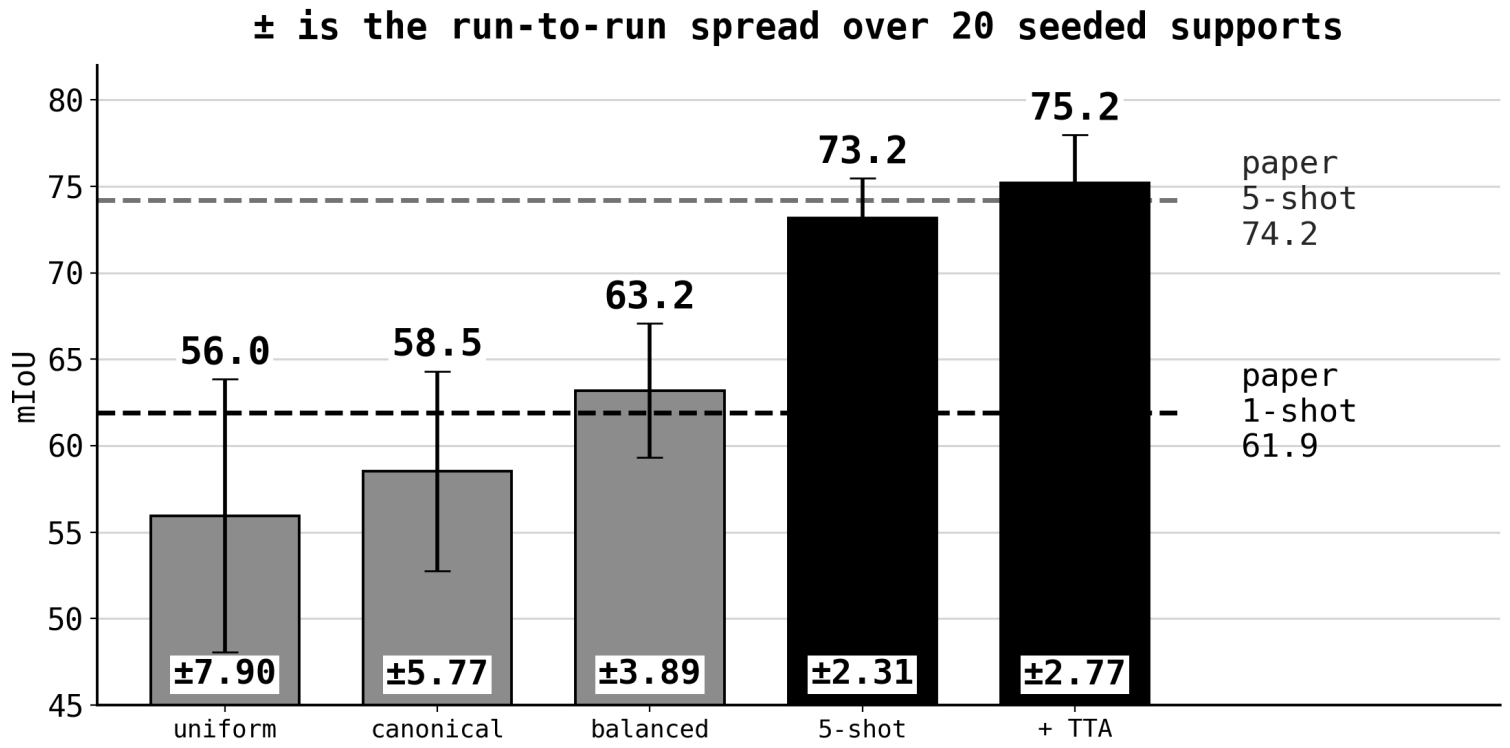
the filter keeps 10/10 of one species and 4/10 of three others

20 draws over 12 species: equal probability is not equal representation



run r starts at species r mod 12

What those two bought



+7.24 mIoU (+12.9%) **head 43.6 → 55.9** **spread 7.90 → 3.89**

TECHNIQUE 3 · SUPPORT-SIDE

Placement curation

Butterflies are not mis-oriented

Pinned wings are already canonical.

They lie loosely on a sheet, so placement varies rather than angle.

Change the statistic, keep the rule

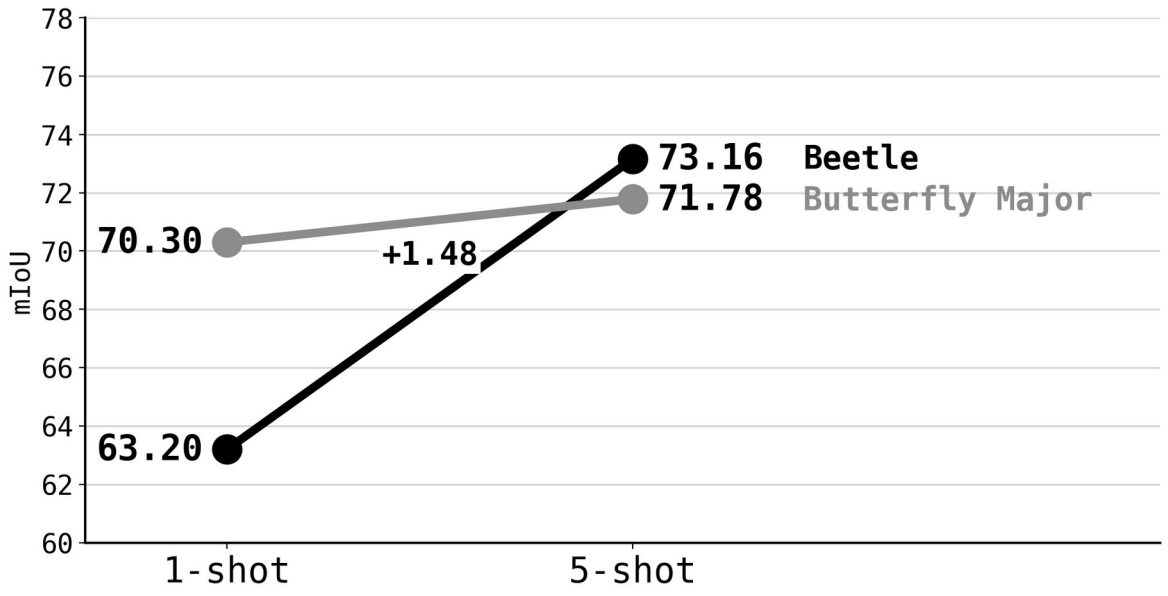
Centroid displacement from the train median.

Threshold = mean + 1 std, the same discipline.

68.54 → 70.30 **+1.90 (+2.8%)**

the two criteria fired on disjoint subspecies, each where its diagnostic predicted

More references

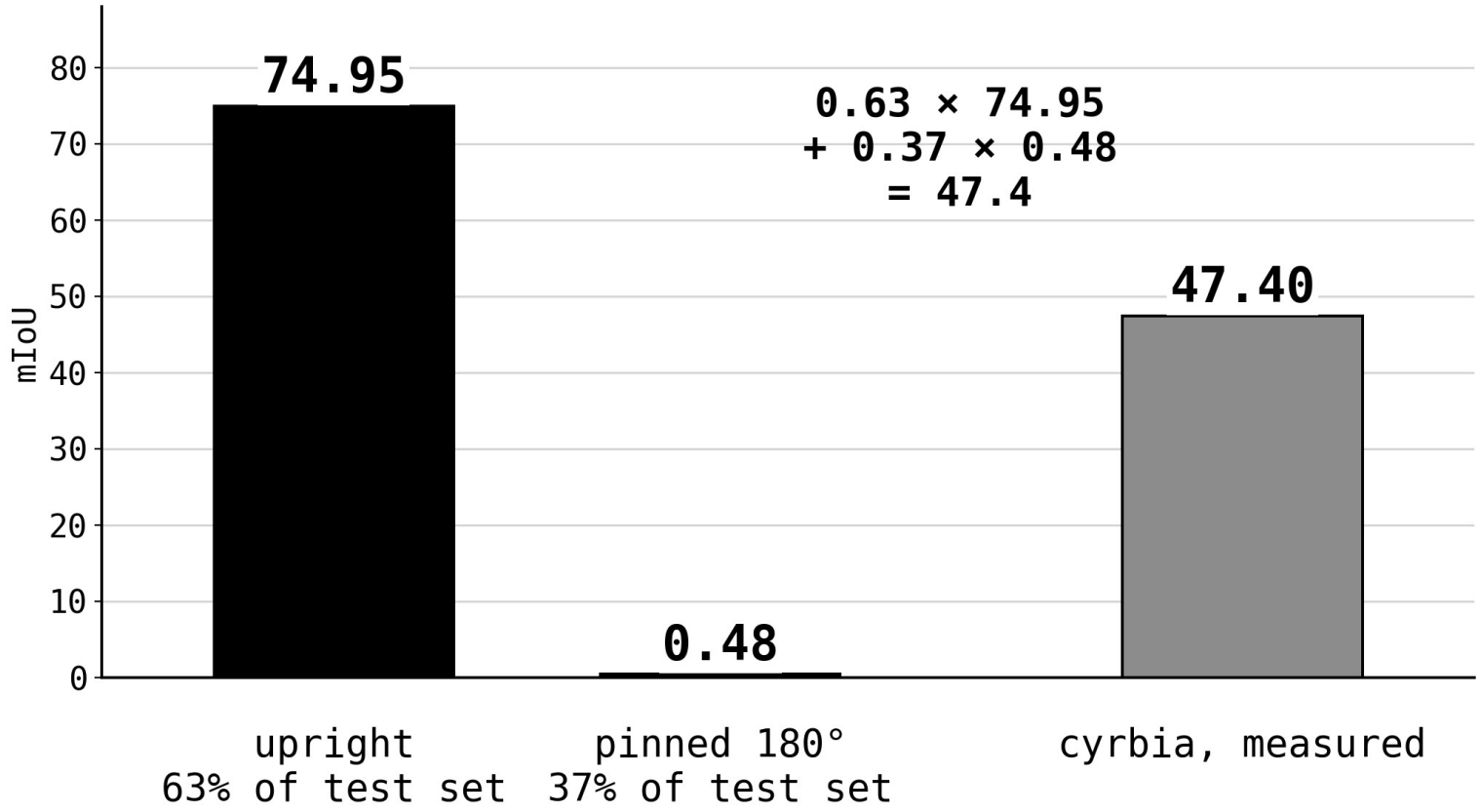


beetle +9.96 (+15.8%) **butterfly +1.48 (+2.1%)**

butterfly gains 1.48; its defect is on the query side

WHY SUPPORT-SIDE STOPS

The query-side defect

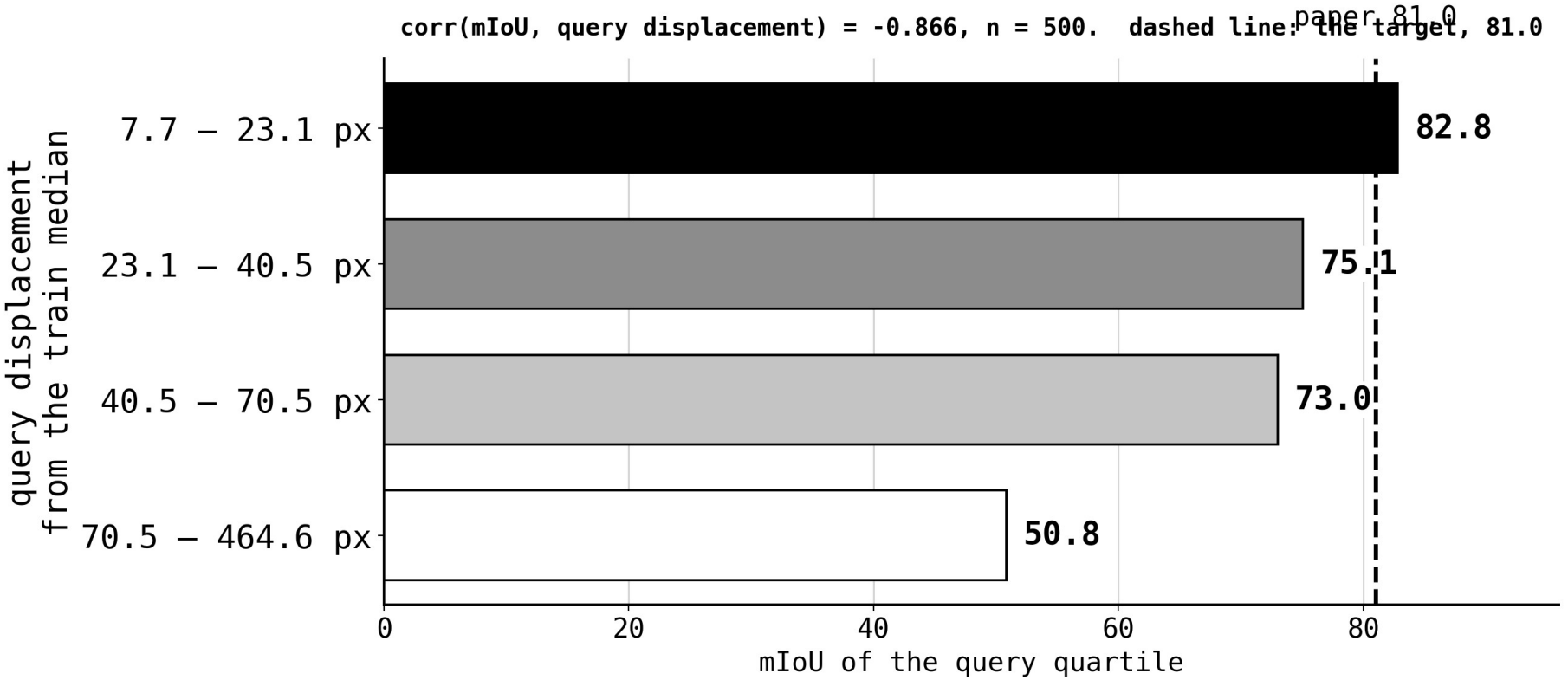


0.00 is what the copy baseline scores on those specimens, with no model involved

curation +3.56 · five shots +2.85 · the level did not move

THE GENERAL FORM OF IT

Query arrangement decides the score



the top quartile already beats the target; the deficit is in the tail

TECHNIQUE 5

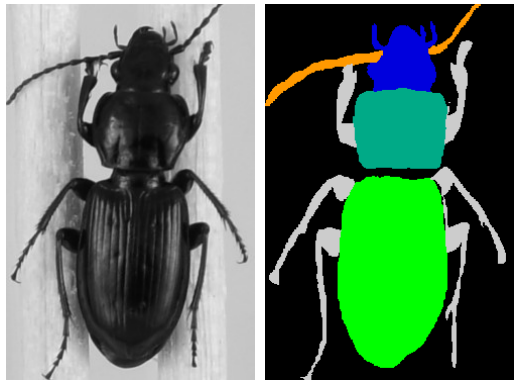
ORIENTATION TTA

The largest single gain in the project, and the least machinery.

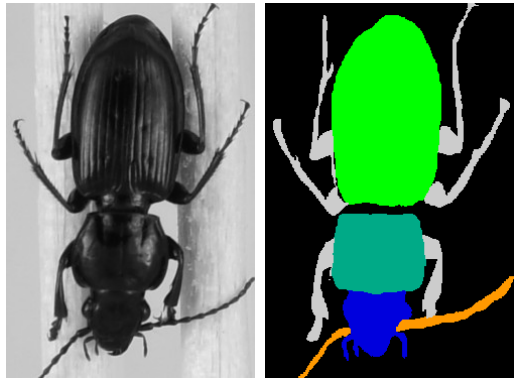
No training and no extra labels.

TTA, how it works: 1 of 2

pass 1, as given



pass 2, support rotated 180°



the query, never touched



01 **Image and mask rotate together**

pixel-aligned, or the mask is useless

02 **Only the support rotates**

its orientation relative to the query is what matters

03 **The rotated variant is cached apart**

@r180, or pass 2 gets the wrong embeddings

TTA, how it works: 2 of 2

```
top = logits.max(axis=0)          # best trait score per pixel
fg  = top > 0                     # the pixels it claims
confidence = top[fg].mean()       # −inf if it claims nothing
```

mismatched orientation → scattered and weak

matching orientation → compact and strong

No ground truth is consulted

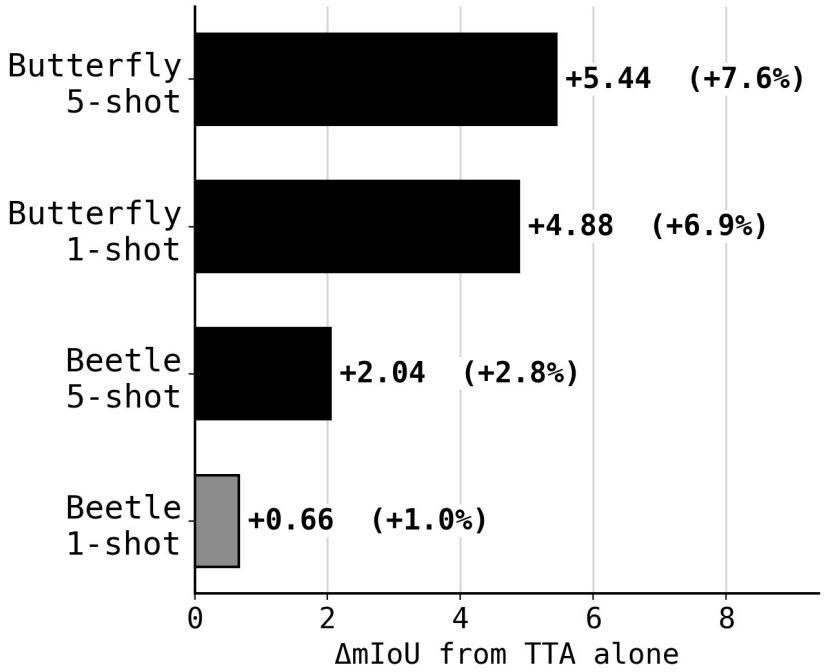
the choice comes from logits alone

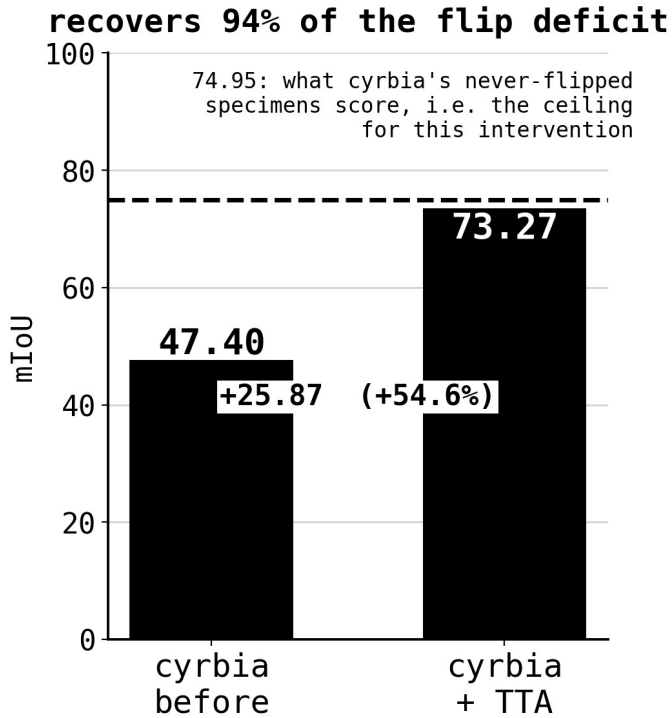
2× inference, not 10×

both orientations cached, so pass 2 is a hit

it chooses between two predictions; it cannot improve either one

What TTA bought



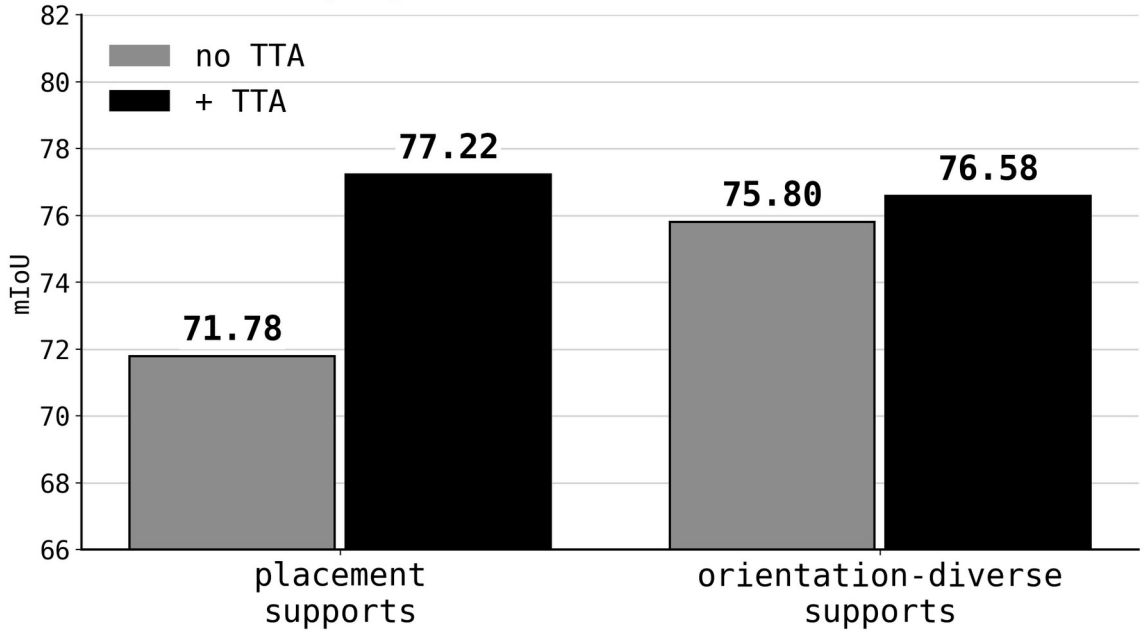


cyrbia 47.40 → 73.27 (+54.6%). cost: its spread rises 6.81 → 10.89

TECHNIQUE 6, AND A WARNING

Two fixes, one defect

support-side +4.02 · query-side +5.44 · both +4.80 · interaction −4.66



technique 6 is support-side: the five references span both orientations, and the memory bank selects rather than blends

best arm: TTA on placement-curated references, 77.22 (+7.6%)

TECHNIQUE 7

CHANGING THE LOSS

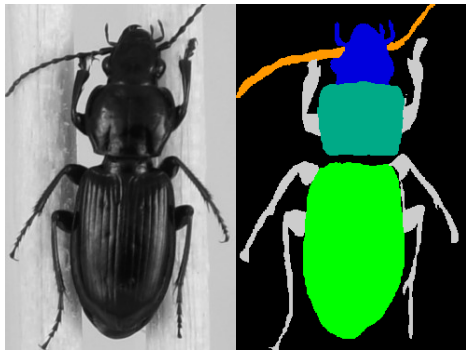
OC-CCL, a cycle-consistency fine-tune.

The most machinery in the project.

HOW IT WORKS

The cycle, mechanically

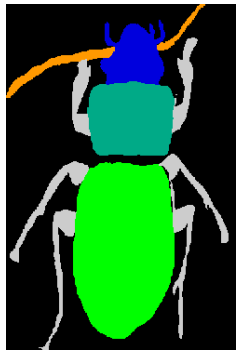
1 / labeled reference



2 / propagate to the unlabeled query



3 / prompt back, reconstruct



loss = BCE + Dice between the reconstruction and the reference's own labels

no new labels required; the reference's own mask is the target

TWO THINGS IT NEEDS TO BE ABLE TO LEARN

Two preconditions

1 · the closing prompt

(pred > 0).float()

severs the graph, so nothing downstream reaches the forward leg.

(logits + 10) / 20

same logits, linear path.

forward ÷ reverse gradient: 0.00 → 26×, and 351× per-trait.

2 · what it supervises

One merged channel supervises the silhouette.

C trait channels supervise assignment.

Each pays where its own objective has room.

butterfly silhouette had room → +0.85 to +1.25

beetle silhouette at 90.8 → no effect

A TERM BORROWED FROM ANOTHER PAPER

SCNP

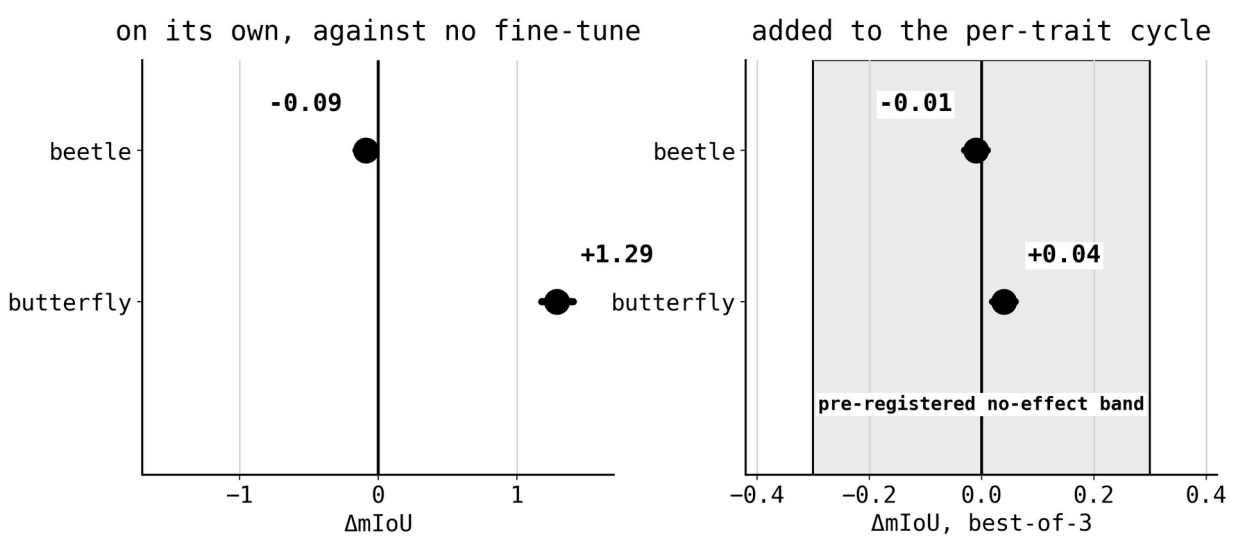
the transform

same class → min-pool the neighbours

other class → max-pool the neighbours

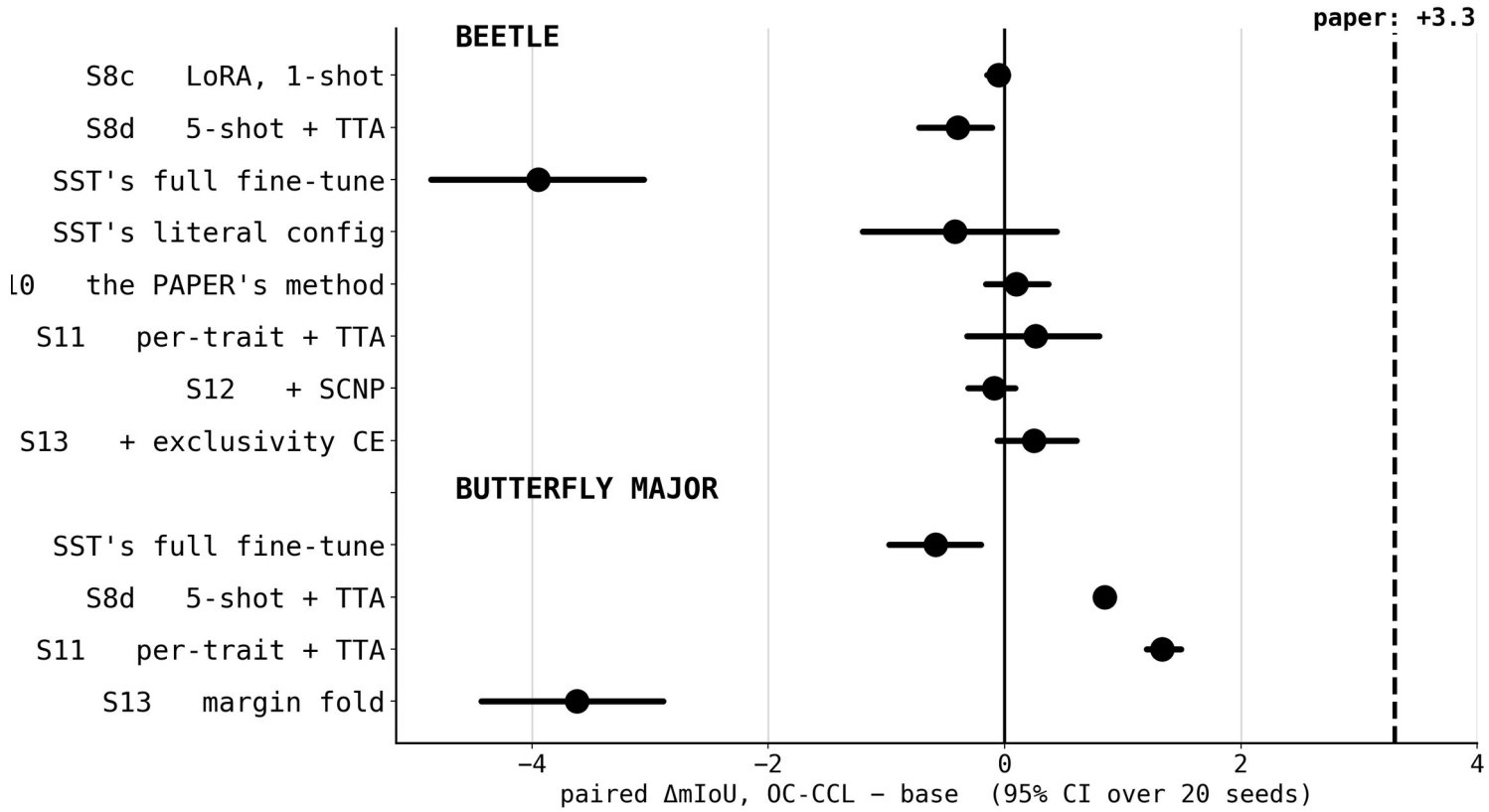
why it should have paid here

an isolated false positive is the worst neighbour of up to nine pixels



tried twice: on the merged silhouette, then on the per-trait cycle where “same class” means same trait

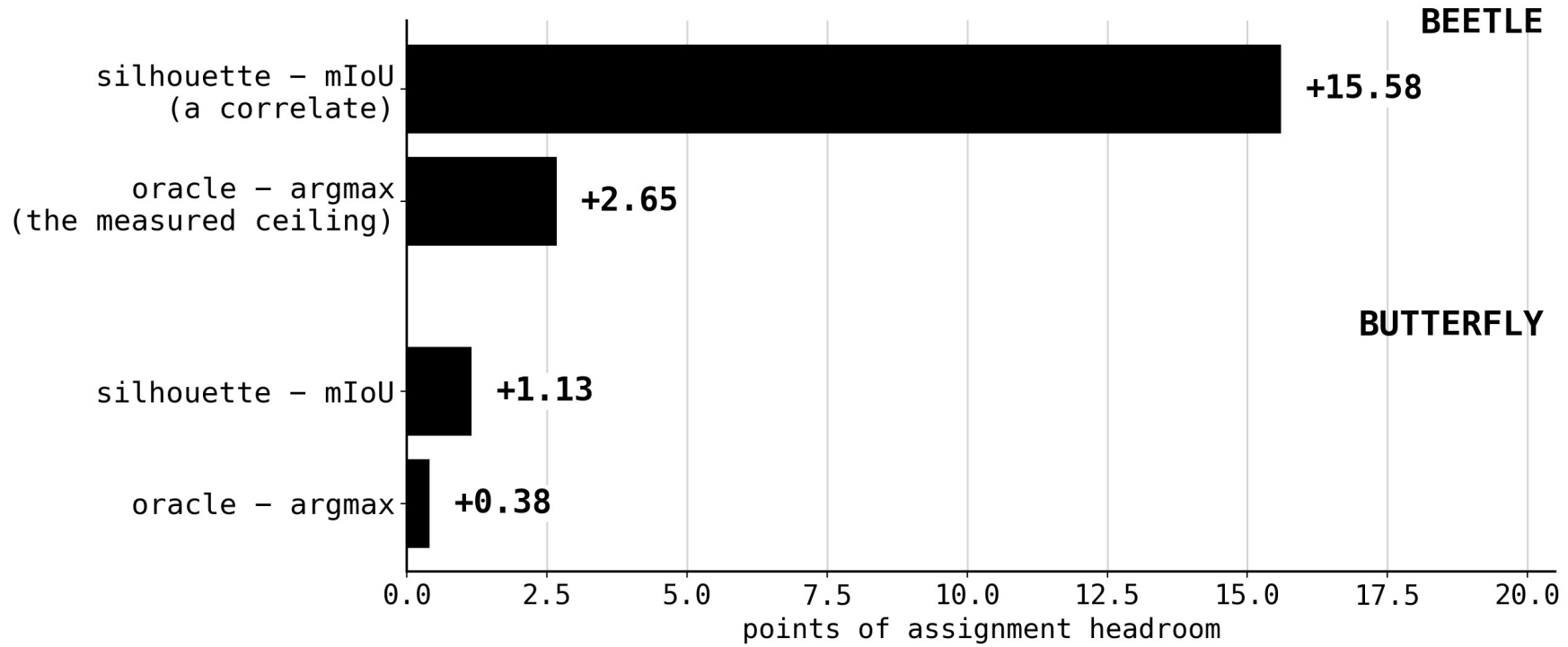
Every loss variant



LoRA and full fine-tuning · 1 and 10 epochs · five learning rates · five loss terms · two datasets

WHY

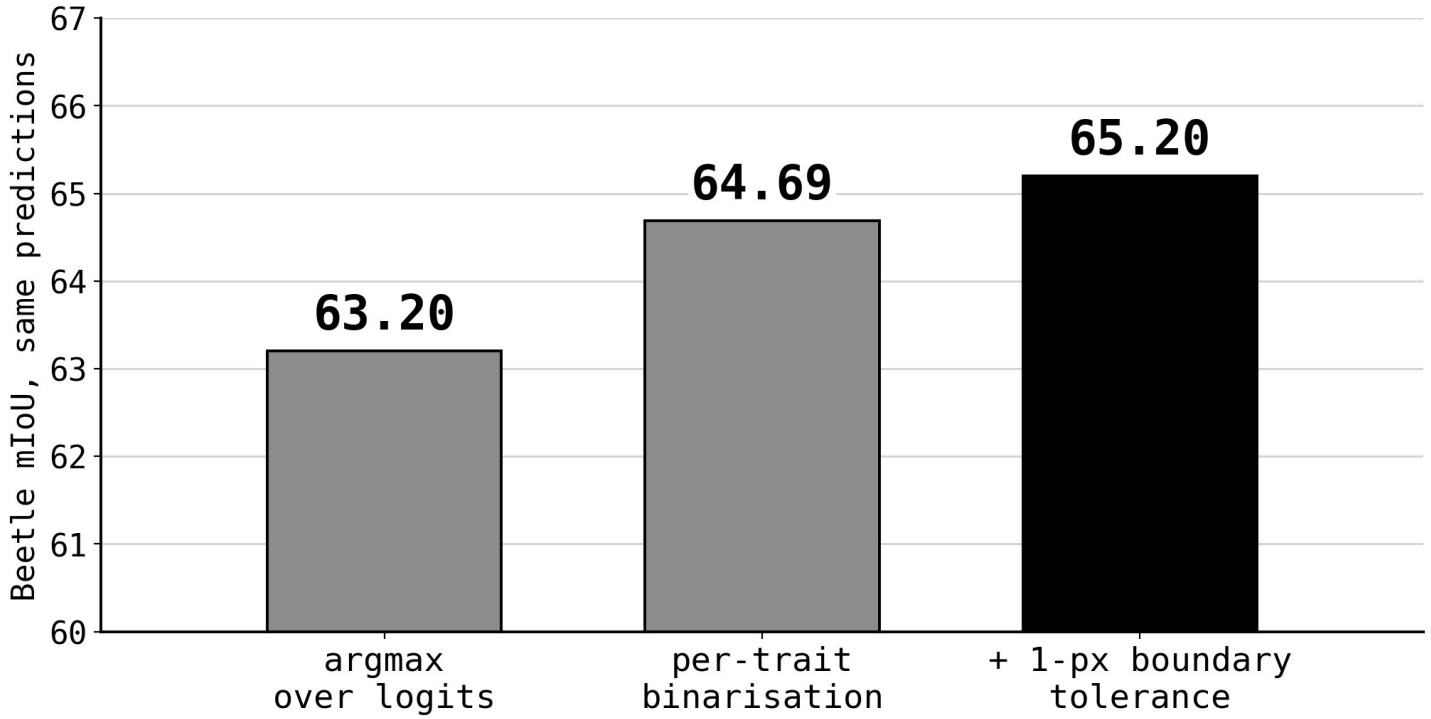
The pot they competed for



the rest is regions the tracker never proposes, which no loss on proposed logits reaches

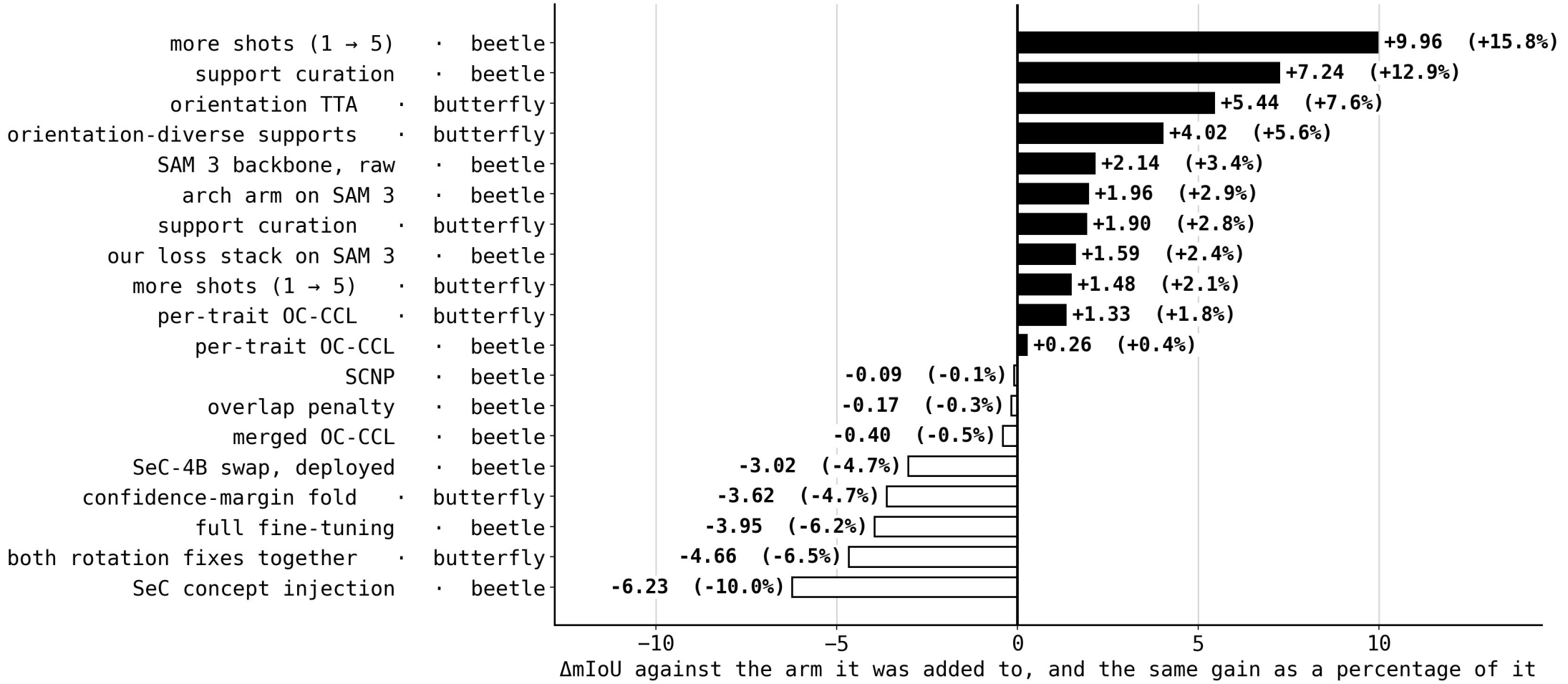
AND ONE THAT ISN'T A MODEL CHANGE AT ALL

Three defensible scorings

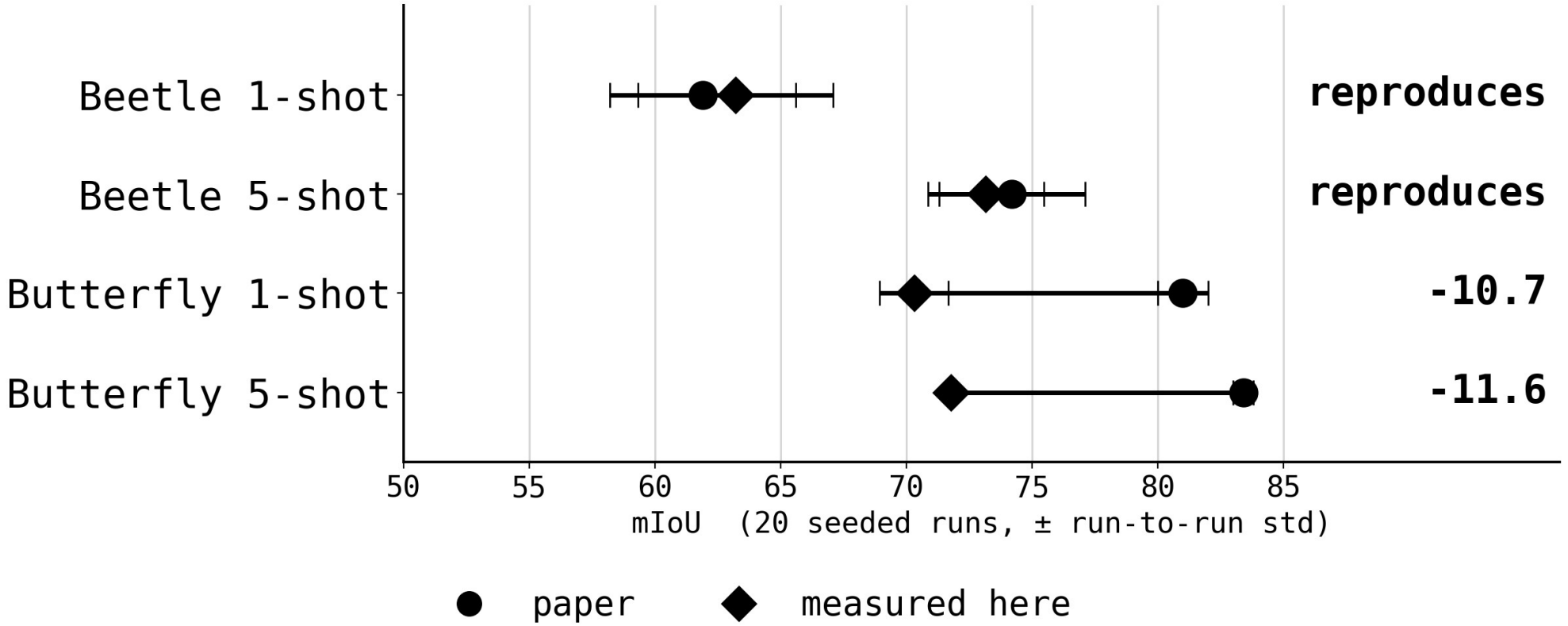


every fine-tune delta in this talk moved by less than 0.3 under all three readings

The ledger


more shots (1 → 5) · beetle
support curation · beetle
orientation TTA · butterfly
orientation-diverse supports · butterfly
SAM 3 backbone, raw · beetle
arch arm on SAM 3 · beetle
support curation · butterfly
our loss stack on SAM 3 · beetle
more shots (1 → 5) · butterfly
per-trait OC-CCL · butterfly
per-trait OC-CCL · beetle
SCNP · beetle
overlap penalty · beetle
merged OC-CCL · beetle
SeC-4B swap, deployed · beetle
confidence-margin fold · butterfly
full fine-tuning · beetle
both rotation fixes together · butterfly
SeC concept injection · beetle
+9.96 (+15.8%)
+7.24 (+12.9%)
+5.44 (+7.6%)
+4.02 (+5.6%)
+2.14 (+3.4%)
+1.96 (+2.9%)
+1.90 (+2.8%)
+1.59 (+2.4%)
+1.48 (+2.1%)
+1.33 (+1.8%)
+0.26 (+0.4%)
-0.09 (-0.1%)
-0.17 (-0.3%)
-0.40 (-0.5%)
-3.02 (-4.7%)
-3.62 (-4.7%)
-3.95 (-6.2%)
-4.66 (-6.5%)
-6.23 (-10.0%)
−10
−5
0
5
10
ΔmIoU against the arm it was added to, and the same gain as a percentage of it

Where the numbers land



~5 of butterfly's last 6 points sit below the silhouette ceiling

The two swaps

SeC-4B

A concept LVLM steering a retrained SAM 2.1 tracker: identity reasoning, not pixel memory.

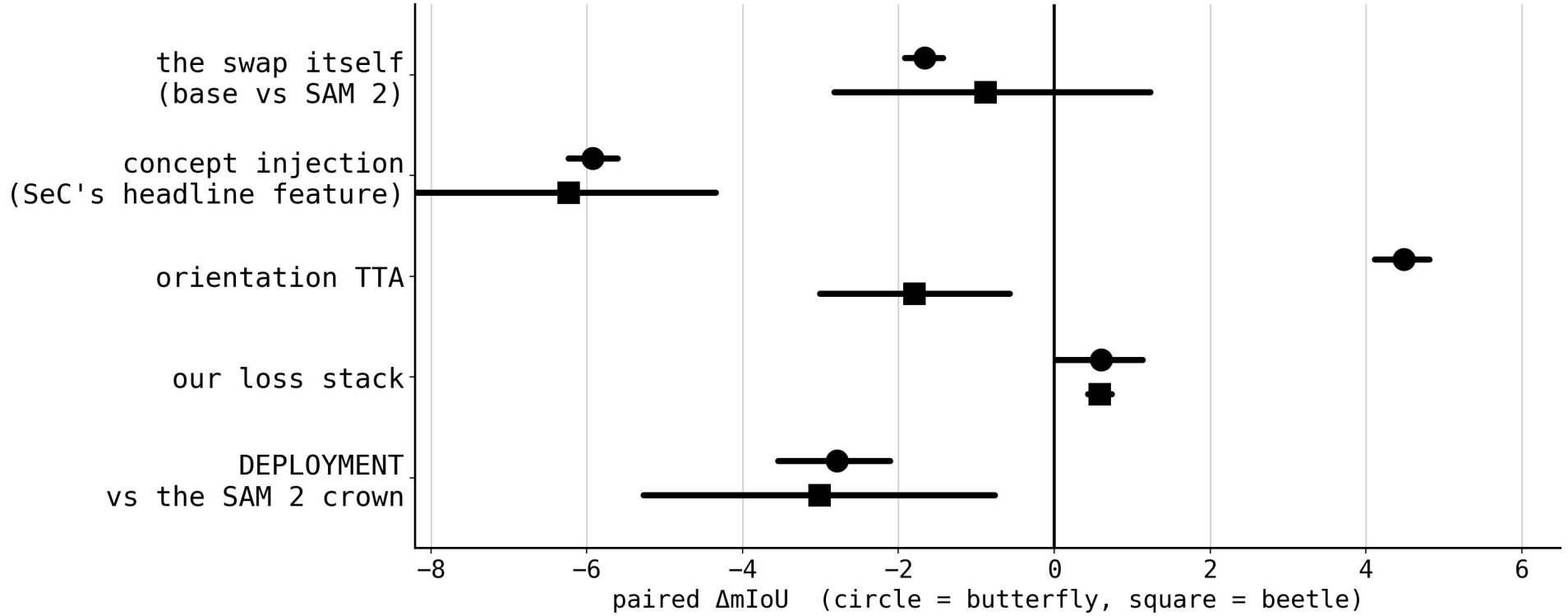
SAM 3

The same tracker design one generation on, so every technique here ports unchanged.

same 20 draws · same protocol · the winning stack on top

one new arm: LoRA on the memory attention, the module that forms proposals

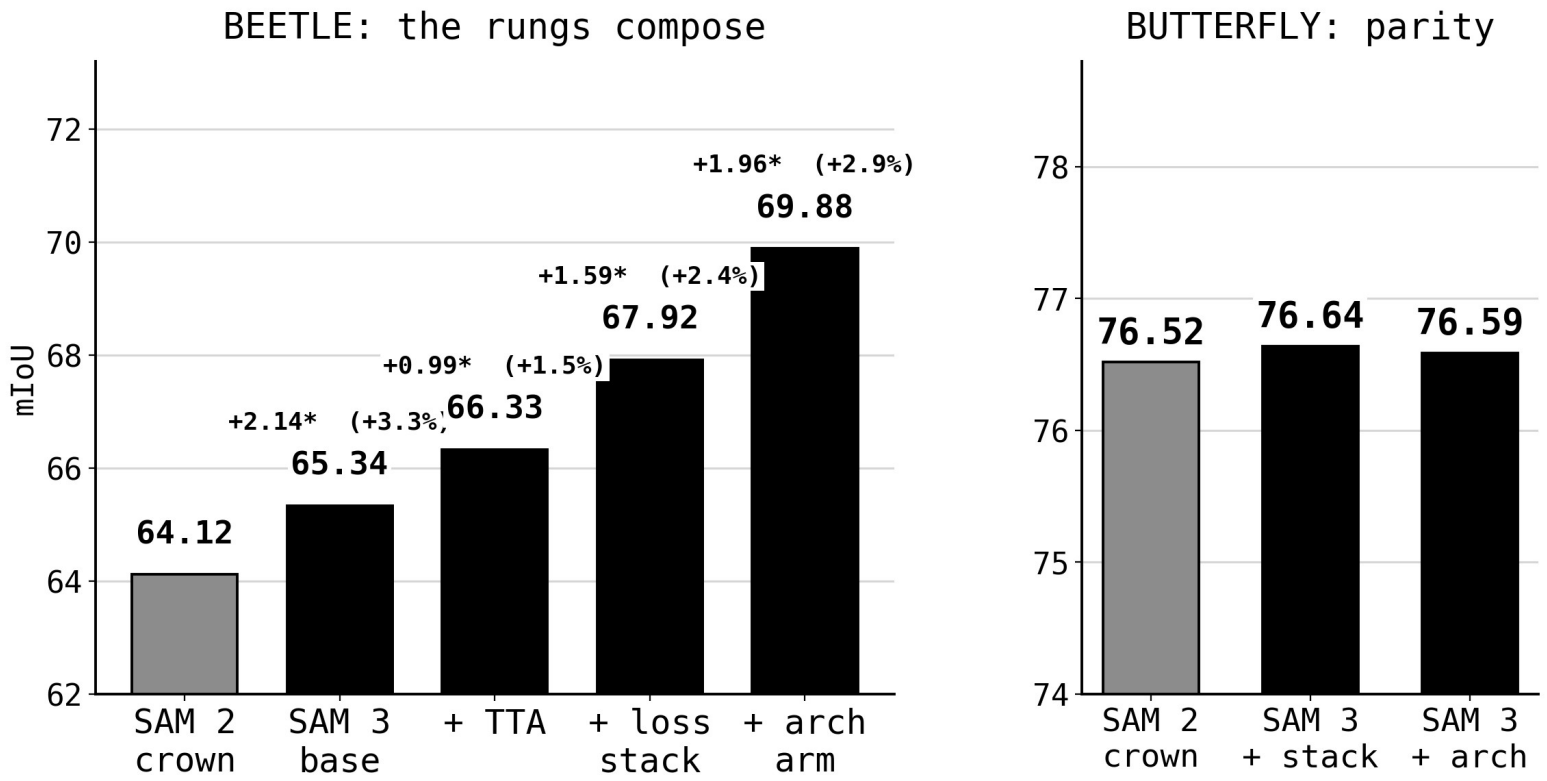
SeC-4B: the swap does not pay



concept injection costs 5.92 and 6.23 mIoU, which is 8.6% and 10.0% of what it was added to

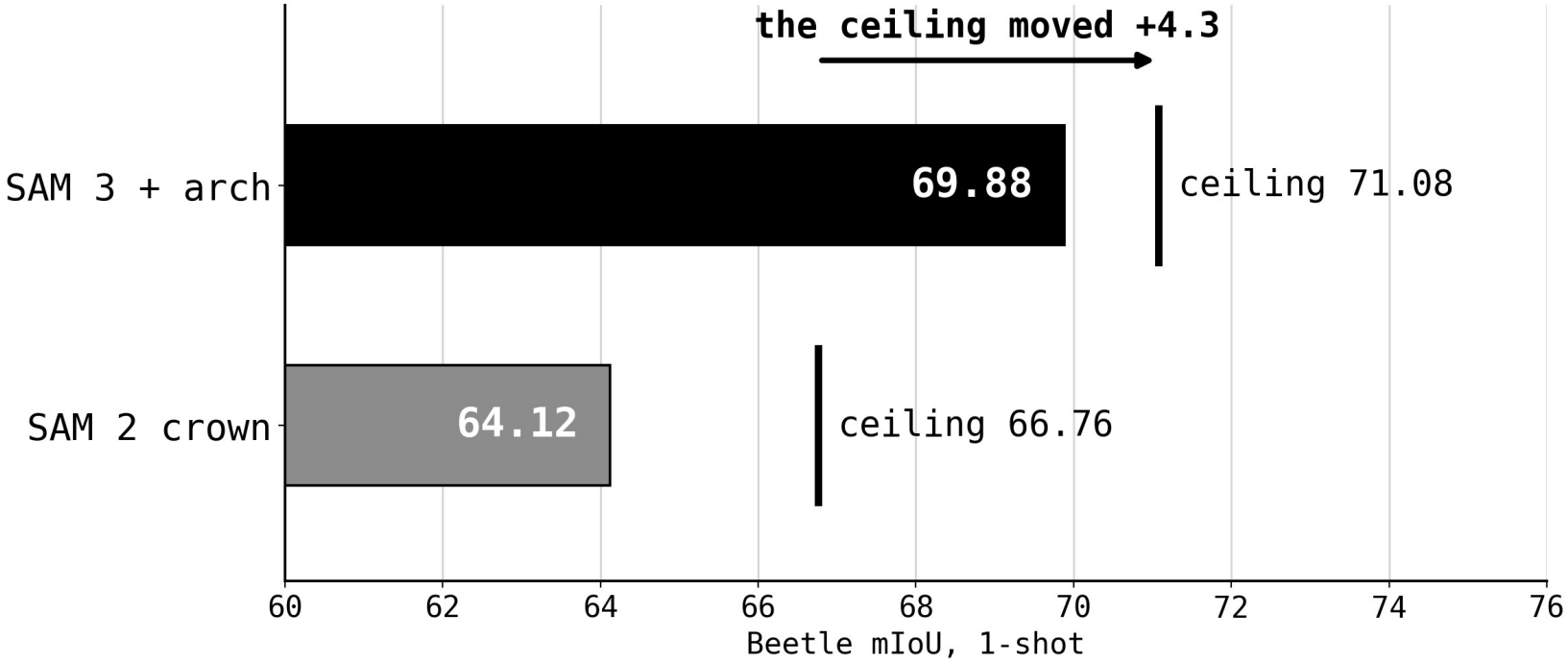
the loss stack still pays on it: +0.60 (+0.8%) and +0.58 (+1.0%), above its value on SAM 2

SAM 3: the swap pays



beetle's best 1-shot number: 64.12 → 69.88, +5.77 ∗ (+9.0%) vs the SAM 2 stack

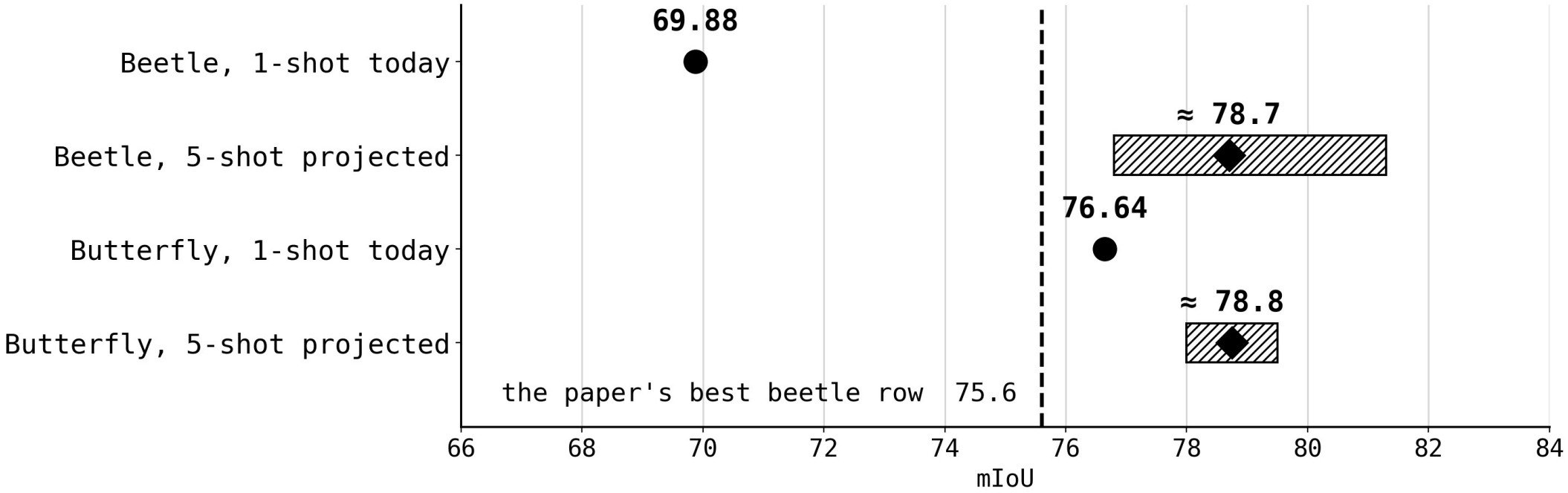
The ceiling itself moved



the arch arm trains the memory attention, the module that decides what is proposed

so it reaches error mass that a loss on proposed logits cannot

What is pre-registered next



bands assembled from measured components: shot lift, TTA's growth with shot, the stack, the arch arm

a central landing sits ≈ +3 over the paper's best beetle row

IN ONE LINE EACH

Where it lands

On SAM 2, every technique that paid changed what the model is shown, not what it optimises.

Orientation TTA is the largest inference-time gain: +5.44, or +7.6%, label-free and untrained.

On SAM 3 that ordering reverses: the loss stack and the arch arm are +3.55 of beetle's +5.77, which is +9.0%.

They moved the ceiling rather than the assignment. Butterfly's residual survived both swaps.

Canonicalization, implemented

orientation

body_axis_angle(lm, head, tail)

atan2(tail_cy - head_cy,
tail_cx - head_cx)

directed, so a 180° rotation is a different angle, not the same axis

accept if deviation <= circular_std(train)

placement

trait_centroids(lm, expected)

reference = per-trait MEDIAN over the train split

displacement = mean distance over the traits both configs share

accept if disp <= train mean + 1 std

completeness

len(trait_ids(lm)) == expected

the 180° test, used for strata: orientation_of() compares against the reference and the reference mirrored about the image centre, at the mask's own extent

circular_std = sqrt(-2 ln R), the train pool's own spread

every threshold is recorded per run and pinned by a test

From pool to draw

canonical_pool(ds, group, criteria)	reads TRAIN masks only, returns (pool_ids, stats); stats carry the thresholds and the rejection counts into the bundle
draw_supports(ds, group, pool, selection, shot, run)	the single dispatch point; three call sites had drifted before it existed
derive_seed(dataset, group, shot=0, run)	blake2b of the key string, no global RNG; shot is pinned to 0 so 1-shot is a strict prefix of 5-shot
stratified_supports(...)	run r takes stratum (r + k) mod S, member drawn by the run's seed; systematic coverage rather than equal probability
orientation_diverse_supports(...)	n_rot = round(shot × rotated fraction of the train pool), clamped to [1, shot-1]; degenerates to the ordinary draw where one orientation exists
support_set_hash(resolved ids)	sha1 of the IDs actually drawn per run; two arms may only be paired if this matches

memoised above canonical_pool, never inside it, so the leakage test cannot pass vacuously

What keeps it honest

- threshold is the a priori train circular std

  so it cannot be retuned

- pool is unchanged when the test split is perturbed

  no leakage

- orientation is measured against the mask's own extent

  beetle crops are 172 to 431 px

- balanced draws preserve prefix nesting

  1-shot vs 5-shot stays paired

- the support draw is dispatched in exactly one place

  the test discovers callers rather than listing them

- registry refuses to pair two different support arms

  protocol.support_selection is a comparability field

recorded per run: criteria, criteria_applied, pool_size, train_size, rejections by reason, threshold_deg, train_circular_mean_deg, train_resultant_R, threshold_rule, placement_threshold_px

the approximation is auditable after the fact, which is the point